

Simulation-to-Reality Correlation Standard - (SRCS)

OOF™ Origin Open Foundation™

Independent Methodological Authority

Simulation-to-Reality Correlation Standard (SRCS) defines the governance conditions under which materially relevant simulated conditions, behaviors, states, relationships, events, transitions, and outputs are systematically compared with corresponding Operational Reality in order to establish where meaningful correlation exists, where material deviation occurs, how alignment varies across conditions and dimensions, and what uncertainty and limitations bound any resulting Simulation-to-Reality Correlation claim.

SRCS establishes the governed relationship between:

what the simulation produces

and:

what corresponding reality demonstrates.

It does not assume that similarity establishes equivalence.

It does not assume that one successful comparison establishes general correspondence.

It does not convert observed agreement into Validation State.

Instead, SRCS provides the methodological correlation layer through which similarities and differences between Simulation and Operational Reality remain explicit, bounded, condition-specific, traceable, and available for subsequent validation, reliance, recalibration, or simulation lifecycle governance.

Foundational Distinction

SIMULOS® establishes:

Simulation Similarity ≠ Reality Equivalence

A simulation may resemble Operational Reality closely in one dimension while diverging materially in another.

It may correlate strongly: under nominal conditions,

and weakly: near boundaries, during transitions, under stress, in rare events, or when multiple conditions interact.

Therefore Simulation-to-Reality Correlation must remain multidimensional,

contextual, and bounded.

Purpose

The purpose of SRCS is to prevent simulation programs from moving directly from:

“The simulation produced this.”

to:

“Reality behaves this way.”

without a governed correlation process between those statements.

SRCS exists because: a realistic-looking output may still be materially wrong, aggregate agreement may conceal local deviation, one accurate subsystem may coexist with another poorly represented subsystem, correct nominal behavior may coexist with incorrect failure behavior, apparent statistical similarity may conceal different causal or dynamic structures, correlation may deteriorate over time, reality itself may change, and simulation agreement may depend upon conditions narrower than the resulting claim implies.

Architectural Position

SRCS is the tenth Parent Standard of SIMULOS® — Simulation Governance Architecture.

Its architectural position follows:

Simulation Execution

↓

Simulation Output Governance

↓

Simulation-to-Reality Correlation

and precedes downstream SIMULOS® lifecycle governance concerning the legitimacy, continued applicability, reliance, change, and future use of simulation results.

SRCS is therefore the point at which SIMULOS® asks:

How does the governed simulated result relate to the actual reality it claims to represent?

Governance Flow Position

Governed Simulation Outputs

↓

Corresponding Operational Reality Basis

↓

Simulation–Reality Comparison

↓

Correlation Criteria

↓

Correlation Assessment

↓

Deviation Identification

↓

Reality Alignment Assessment

↓

Correlation Uncertainty

↓

Bounded Simulation-to-Reality Correlation

↓

Downstream Validation / Reliance / Recalibration / Lifecycle Governance

Operational Space

SRCS governs: Simulation-to-Reality comparison, comparison-object identity, reality reference identity, correlation target definition, comparison dimension selection, correlation criteria, correspondence thresholds, behavioral correspondence, state correspondence, event correspondence, transition correspondence, output correspondence, distributional correspondence, temporal correspondence, spatial correspondence, sequence correspondence, interaction correspondence, boundary correspondence, stress-condition correspondence, recovery correspondence, comparative context, correlation strength, conditional correlation, partial correlation, local correlation, global correlation, correlation asymmetry, deviation identification, deviation magnitude, deviation direction, deviation localization, deviation persistence, deviation pattern, systematic deviation,

random deviation, reality alignment, alignment boundaries, alignment deterioration, alignment improvement, correlation uncertainty, correlation limitations, correlation scope, correlation conditions, correlation change, correlation traceability, and Simulation-to-Reality Correlation records.

Operational Scope

SRCS applies wherever a simulation explicitly or implicitly claims to represent aspects of reality sufficiently for comparison.

It may apply across: aerospace, spacecraft and mission simulation, autonomous systems, robotics, AI systems, digital twins, industrial processes, manufacturing systems, transportation, infrastructure, energy systems, environmental simulation, scientific models, financial systems, healthcare systems, human-system interaction, multi-agent systems, cyber-physical systems, operational planning, and future simulation environments.

SRCS is technology-neutral.

It does not prescribe one mathematical correlation coefficient, statistical method, physical measurement system, calibration method, or validation technique.

The governed question remains:

Does the simulation correspond to reality sufficiently for the specific relationship being claimed?

Correlation Object

A Correlation Object is the specific simulation behavior, state, output, event, relationship, transition, or other simulation result being compared with a corresponding reality reference.

Examples may conceptually include:

Simulated trajectory ↔ observed trajectory

Simulated thermal behavior ↔ measured thermal behavior

Simulated agent response ↔ observed agent response

Simulated failure progression ↔ operational failure progression

Simulated demand pattern ↔ actual demand pattern

Simulated system transition ↔ observed system transition

SRCS requires clarity about what exactly is being correlated.

Reality Reference

A Simulation-to-Reality Correlation requires a corresponding Reality Reference.

A Reality Reference may consist of: operational measurements, sensor records, physical observations, mission telemetry, field behavior, experimental observations, operational incidents, verified historical records, or another sufficiently identified representation of Operational Reality.

SRCS does not govern Operational Reality itself.

ORA™ may provide or govern the relevant Operational Reality basis.

SRCS governs the relationship between that basis and the simulation.

Simulation-to-Reality Correlation Basis

A governed correlation should preserve the relationship:

Simulation Object

↓

Simulation Configuration

↓

Scenario

↓

Simulation Execution

↓

Simulation Output / Behavior

↔

Corresponding Reality Condition

↓

Operational Observation

↓

Comparison



Correlation Assessment

This creates a traceable bridge rather than an unsupported statement of realism.

Comparable Conditions Principle

Simulation and reality should not be compared as if they were equivalent where materially relevant conditions differ.

Possible comparison dimensions may include: object version, environment, input conditions, initial state, operating state, system configuration, population, timing, scenario, boundary conditions, stress level, and dependencies.

SIMULOS® establishes:

Correlation requires sufficient comparability of the conditions being compared.

Correlation Target

A Correlation Target defines what correspondence is being examined.

For example:

Trajectory correspondence

is different from:

energy-consumption correspondence

which is different from:

failure-sequence correspondence.

Strong correlation in one target must not be generalized automatically to another.

No Correlation Transfer Principle

SIMULOS® establishes:

Correlation demonstrated for one behavior, condition, subsystem, environment, population, or scenario does not automatically transfer to materially different ones.

Correlation Criteria

Every material Simulation-to-Reality Correlation assessment should operate against sufficiently defined criteria.

Correlation Criteria may concern: allowable deviation, behavioral pattern, directionality, ordering, timing, distribution, state transition, physical

tolerance, response shape, event occurrence, frequency, magnitude, or another simulation-specific relationship.

Criteria should be defined sufficiently to distinguish:

acceptable correspondence

from:

material deviation. Correlation Criteria $\neq$ Validation Criteria

Correlation Criteria answer:

How similar or aligned are the Simulation and Reality with respect to this defined relationship?

VALIDOS® Validation Criteria may later answer:

Is the resulting correlation evidence sufficient to support a particular Validation Determination?

These are related but distinct governance questions.

Direct Correlation

Direct Correlation occurs where a simulated quantity or behavior can be compared directly with a corresponding observed quantity or behavior.

Conceptually:

Simulated X $\leftrightarrow$ Observed X

Direct correspondence may be strong, weak, conditional, or absent.

Indirect Correlation

Some simulated properties cannot be observed directly.

Correlation may therefore depend upon: proxies, derived observations, intermediate measurements, aggregate behavior, or indirect operational signals.

Where correlation is indirect, the intermediate relationship should remain explicit.

Proxy Correlation Principle

SIMULOS® establishes:

Agreement with a proxy does not automatically establish equivalence with the underlying reality the proxy represents.

Behavioral Correlation

Behavioral Correlation assesses whether simulated behavior sufficiently corresponds to observed behavior.

This may include: response to inputs, adaptation, recovery, escalation, coordination, failure, control behavior, or agent interaction.

State Correlation

State Correlation assesses whether the simulated system reaches or occupies states corresponding sufficiently to observed states.

Transition Correlation

Transition Correlation assesses whether the simulation reproduces materially relevant movement between states.

This is important because:

Correct State A + Correct State B $\neq$ Correct A $\rightarrow$ B Transition Temporal Correlation

Temporal Correlation concerns whether simulation events occur: at comparable times, with comparable delays, in comparable sequences, or across comparable durations.

A simulation may reproduce an event but still materially misrepresent when it occurs.

Spatial Correlation

Spatial Correlation concerns whether simulated positions, distributions, paths, fields, or spatial relationships correspond sufficiently to reality.

Distributional Correlation

Aggregate values may agree while distributions differ.

For example:

same mean

may coexist with:

different variance, tails, clustering, or multimodality.

SRCS therefore does not allow aggregate similarity automatically to establish distributional correlation.

Sequence Correlation

Systems may generate the correct events in the wrong order.

Where order matters, SRCS evaluates:

Event Identity + Event Sequence

rather than event presence alone.

Interaction Correlation

Interaction Correlation concerns whether simulated relationships between: agents, subsystems, humans, environments, dependencies, or other actors

correspond sufficiently to observed relationships.

Boundary Correlation

A simulation may correlate well in the center of the operating envelope but poorly near limits.

SRCS therefore supports correlation assessment across:

Nominal Region

Boundary Region

Threshold Region

and where relevant:

Beyond-Nominal Region.

Stress Correlation

Where Operational Reality provides evidence from degraded or stressed conditions, SRCS may compare:

Simulated Stress Behavior

with:

Observed Stress Behavior.

Strong nominal correlation must not substitute for absent stress correlation where stress behavior materially matters.

Recovery Correlation

Recovery behavior may require separate correlation.

A model may accurately reproduce failure onset while poorly reproducing: fallback, restoration, adaptation, restart, or recovery duration.

Local Correlation

Local Correlation exists where Simulation and Reality correspond sufficiently within a bounded region of: state, environment, population, configuration, parameter space, or scenario conditions.

Global Correlation

Global Correlation represents broader correspondence across the declared Simulation Scope.

Global correlation should not be claimed by extrapolating from narrow local correlation without an adequate basis.

No Local-to-Global Inflation Principle

SIMULOS® establishes:

Strong local Simulation-to-Reality Correlation does not establish global correlation across unexamined Scenario Space.

Conditional Correlation

Correlation may hold only under explicit conditions.

For example:

Strong correlation under nominal thermal conditions

but:

weak correlation under rapid thermal transition.

The resulting correlation record should preserve those conditions.

Partial Correlation

Different aspects of the same simulation may correlate differently.

For example:

Position — Strong

Velocity — Strong

Energy Consumption — Moderate

Recovery Timing — Weak

The simulation should not be given one undifferentiated correlation label if material dimensions differ.

Correlation Profile

A Simulation-to-Reality Correlation Profile may represent multidimensional correspondence across relevant dimensions.

This is often methodologically stronger than one global score.

No Single-Score Requirement

SIMULOS® does not require correlation to be represented through one numerical metric.

A single score may conceal: strong and weak dimensions, local deviation, conditional relationships, failure-region divergence, or uncertainty.

Where a numerical score is used, its meaning and limitations should remain explicit.

Correlation Strength

Where useful, correlation may be classified as:

Strong

Moderate

Weak

Conditional

Partial

No Material Correlation

or:

Correlation Undetermined

provided the classification is tied to explicit criteria. Correlation Strength
≠ Validity

SIMULOS® establishes:

Strong Simulation-to-Reality Correlation does not itself establish that the simulation is Validated for every intended use.

Correlation may become one component of a broader VALIDOS® validation record.
Deviation

A Simulation-to-Reality Deviation is a materially relevant difference between simulated behavior and corresponding Operational Reality.

Deviation may concern: value, timing, direction, sequence, magnitude, distribution, state, transition, interaction, failure, recovery, or another governed property.

Deviation Is Not Automatically Failure

A deviation may be: expected, acceptable, immaterial, explainable, model-induced, data-induced, environment-induced, reality-induced, or materially problematic.

SRCS governs identification and characterization before drawing stronger conclusions.

Deviation Magnitude

Deviation Magnitude describes how large the difference is relative to the applicable comparison basis.

Magnitude should be interpreted together with materiality.

A numerically small deviation may be critical near a threshold.

A numerically larger deviation may be inconsequential elsewhere.

Deviation Direction

Deviation may systematically: overestimate, underestimate, anticipate, delay, amplify, suppress, accelerate, or otherwise bias simulated behavior relative to reality.

Direction can reveal patterns that aggregate absolute error may hide.

Systematic Deviation

A Systematic Deviation persists predictably across comparable conditions.

This may signal: model bias, incorrect parameterization, representation weakness, input mismatch, missing interaction, or another simulation limitation.

SRCS identifies the pattern.

The appropriate upstream SIMULOS® governance space determines what requires correction.

Random Deviation

Some deviations may vary without stable direction.

Random deviation may still be material where its magnitude affects the intended Simulation Purpose.

Localized Deviation

A simulation may correlate strongly overall while diverging materially in one bounded region.

This creates a Localized Correlation Gap rather than necessarily a global simulation failure.

Deviation Map

Where appropriate, SRCS may represent:

Scenario Space → Correlation Strength / Deviation

This may reveal regions where the simulation remains strongly reality-aligned and regions where it does not.

Deviation Attribution

Where possible, deviations should be connected to potential contributing factors such as: Reality Representation, Simulation Assumptions, model structure, environment configuration, parameters, inputs, transformation, Scenario Coverage, execution conditions, output interpretation, or changed Operational Reality.

Attribution should remain evidence-bounded.

No Automatic Cause Principle

SIMULOS® establishes:

Observed Simulation-to-Reality deviation identifies a difference; it does not automatically establish why that difference exists.

Reality Alignment

Reality Alignment describes the degree to which the simulation continues to correspond sufficiently to relevant Operational Reality across the declared conditions.

Alignment may be:

Strong

Conditional

Partial

Weak

Misaligned

or:

Undetermined.

Reality Alignment Boundary

A simulation may have a clearly bounded region where alignment is sufficient.

Conceptually:

Aligned Region

↓

Transition Region



Misaligned Region

Identifying this boundary may be more informative than assigning one global realism label.

No Reality-Equivalence Claim

SIMULOS® establishes:

No degree of observed Simulation-to-Reality similarity should be represented as complete Reality Equivalence unless such equivalence has been independently and legitimately established for the exact governed claim.

For complex systems, such universal equivalence may often be impossible to establish.

Correlation Uncertainty

A Simulation-to-Reality Correlation claim itself contains uncertainty.

Possible sources include: measurement uncertainty, operational observation limitations, limited reality data, sparse comparison conditions, input uncertainty, scenario mismatch, temporal mismatch, stochastic behavior, environment differences, system change, and uncertainty in correlation methods.

Correlation Uncertainty Must Remain Attached

A correlation statement should not imply greater certainty than its comparison basis supports.

Conceptually:

Correlation Result + Correlation Uncertainty + Conditions + Scope

should remain connected.

No Correlation Precision Inflation

A precisely calculated correlation metric does not imply equivalent certainty in the reality relationship where the observational basis is limited.

Reality Observation Limitation

Operational Reality may itself be incompletely observable.

For example: internal system states may not be measurable, field observations

may be sparse, critical failures may be rare, measurements may be noisy, mission data may be incomplete, or historical records may lack relevant context.

SRCS preserves this limitation rather than treating reality observations as perfect ground truth by default. Operational Reality $\neq$ Perfect Knowledge of Reality

This distinction is important.

ORA™ may govern the best available Operational Reality representation.

SRCS then correlates the simulation against that governed reality basis.

The limitations of that basis should remain visible.

Correlation Coverage

Simulation-to-Reality Correlation should be assessed relative to the Scenario Space actually represented by Operational Reality observations.

If correlation has been established only under:

Conditions A, B, and C

then the simulation should not claim established correlation under:

D, E, and F

without further basis.

Correlation Coverage Gap

A Correlation Coverage Gap exists where materially relevant Simulation Scenario Space lacks sufficient corresponding Operational Reality evidence for meaningful correlation assessment.

This may occur because: the event has never occurred operationally, the condition cannot safely be reproduced, mission opportunities are limited, observations are incomplete, or future systems do not yet exist operationally.

Uncorrelatable Scenario Space

Some Scenario Space may currently be impossible to correlate directly with reality.

For example: catastrophic spacecraft combinations never observed, future autonomy modes, novel environmental conditions, future infrastructure configurations, or hypothetical emergency states.

The correct governance state may be:

Correlation Not Yet Establishable

rather than:

Correlation Assumed.

No Absence-of-Reality-Data Inference

SIMULOS® establishes:

Where corresponding Operational Reality evidence does not exist, correlation must not be manufactured from simulation evidence alone.

Correlation Change

Simulation-to-Reality Correlation may change even if the simulation itself does not.

Reality may evolve through: system modifications, environmental change, population change, infrastructure change, operational behavior change, software updates, mission changes, or new dependencies.

Correlation Drift

Correlation Drift occurs when previously established correspondence between Simulation and Reality weakens or changes over time.

This may be caused by:

Simulation Change

Reality Change

or:

Both.

No Permanent Correlation Principle

SIMULOS® establishes:

A previously established Simulation-to-Reality Correlation must not be assumed permanent when the simulation or corresponding Operational Reality materially changes.

Correlation Reassessment Trigger

Reassessment may be required when there is material change in: Simulation Object, Reality Representation, Simulation Assumptions, model, environment, parameters, inputs, scenarios, execution, output interpretation, operational environment, object version, population, mission phase, dependency, or available reality evidence.

Correlation Improvement

Correlation may improve after: model refinement, better Reality Representation, improved parameters, better input data, new operational observations, additional scenario coverage, or correction of systematic

deviation.

Improvement should be demonstrated, not presumed.

Correlation Deterioration

New Operational Reality evidence may reveal that previous correspondence was overstated or limited.

SRCS must remain capable of producing a weaker correlation conclusion.

No Positive-Bias Principle

SIMULOS® establishes:

Simulation-to-Reality Correlation governance must remain capable of demonstrating weak correlation, partial correlation, material deviation, or unresolved correspondence.

Simulation Calibration Boundary

Correlation findings may inform simulation calibration.

However:

SRCS does not own general model calibration.

Its role is to identify:

where and how Simulation and Reality correspond or diverge.

The appropriate model, parameter, input, representation, or environment governance space may then address the cause.

No Calibration-to-Correlation Circularity

A simulation tuned using one reality dataset should not automatically be represented as independently correlated against that same dataset without disclosure.

SIMULOS® establishes:

Data used to calibrate the simulation and data used to demonstrate Simulation-to-Reality Correlation should remain distinguishable where independence materially affects the claim.

Correlation Independence

Where a stronger correlation claim requires independent reality evidence, that independence should remain explicit.

This prevents:

fit

from being confused with:

independent correspondence.

Overfitting to Reality Samples

A simulation may reproduce observed cases closely while generalizing poorly to new operational conditions.

SRCS therefore distinguishes:

Correlation with Known Observations

from:

Correlation across materially broader Operational Reality.

Reality Holdout Principle

Where appropriate, previously unused Operational Reality observations may provide stronger evidence of generalizable correlation than observations used to construct or calibrate the simulation.

SRCS does not prescribe a specific holdout method.

It preserves the methodological distinction.

Cross-Environment Correlation

Correlation in one environment must not automatically transfer to another.

Examples:

Earth Laboratory ≠ Orbital Environment

Dry Road ≠ Snow

Factory A ≠ Factory B

Population A ≠ Population B

Low-Traffic Environment ≠ Dense Multi-Agent Environment Cross-Version
Correlation

A correlation conclusion for one Simulation Object or real-system version should not automatically apply to materially changed versions.

Correlation Across Mission Phases

For mission systems, correlation may differ across: launch, cruise, orbital operations, docking, landing, autonomous recovery, safe mode, or other phases.

A global mission-level correlation statement may conceal weak phase-specific correspondence.

Autonomous-System Correlation

Autonomous systems present a particularly important correlation problem.

Their simulated behavior may depend upon: perception, internal state, external agents, communication, timing, memory, changing permissions, human intervention, uncertainty, and adaptive decision paths.

SRCS therefore supports correlation not only of final outputs but of: state transitions, decisions, timing, fallback, escalation, recovery, and interaction patterns.

Space and Mission Correlation

Space systems create a particularly difficult Simulation-to-Reality Correlation environment because: real mission observations are expensive, some events occur only once, catastrophic states may never be directly observed before deployment, space environments can be difficult to reproduce physically, mission phases differ materially, telemetry can be incomplete, and operational configurations evolve.

SRCS therefore does not require impossible reality equivalence.

Instead it provides a governed structure for distinguishing:

what has been correlated

from:

what remains extrapolated, simulated-only, uncertain, or operationally unobserved.

This distinction can be critical where mission decisions depend heavily upon simulation.

Autonomous Vehicle Correlation

An autonomous vehicle simulation may correlate strongly with reality for: ordinary road following, standard braking, common pedestrian behavior, and normal weather.

That does not establish equivalent correlation for: rare interactions, construction zones, degraded sensors, unusual road markings, emergency vehicle behavior, or compound edge conditions.

SRCS keeps the correlation boundary visible.

Digital Twin Correlation

A digital twin may continuously compare:

Simulated State

with:

Operational State.

SRCS may govern: alignment, deviation, temporal lag, drift, systematic mismatch, and correlation deterioration.

A digital twin that ceases to correlate sufficiently with the physical system should not continue to be represented as an equivalent operational representation merely because the software remains active.

Correlation Status

A governed Simulation-to-Reality Correlation assessment may result in statuses such as:

Strongly Correlated

Conditionally Correlated

Partially Correlated

Weakly Correlated

Materially Deviated

Not Correlated

or:

Correlation Undetermined

depending upon the governed criteria and scope.

The status must remain tied to: Correlation Target, Scope, Conditions, Reality Reference, Comparison Basis, and Correlation Uncertainty.

Correlation Status ≠ Simulation State

SRCS does not create a general Validation State.

A correlation status is a governed statement about the Simulation-to-Reality relationship.

VALIDOS® may later determine how that relationship contributes to a formal Validation Determination and Validation State.

Correlation Claim Boundary

Any resulting correlation claim must remain no broader than:

Compared Simulation Conditions Corresponding Operational Reality Conditions
Correlation Criteria Observed Relationship Known Deviation Correlation
Uncertainty Applicable Scope No Upward Correlation Claim Principle

SIMULOS® establishes:

A Simulation-to-Reality Correlation claim must not exceed the strongest correspondence demonstrated by the materially relevant comparison record.

Correlation Traceability

A complete correlation path should remain reconstructable:

Simulation Object

↓

Scenario

↓

Simulation Execution

↓

Simulation Output / Behavior

↓

Correlation Target

↔

Operational Reality Reference

↓

Comparison Criteria

↓

Observed Correspondence

↓

Deviation

↓

Reality Alignment

↓

Correlation Uncertainty

↓

Correlation Status Point-in-Time

Correlation Reconstruction

SRCS should support questions such as:

Which Simulation version was compared?

Against which Operational Reality basis?

Under which conditions?

Which criteria were used?

Which dimensions correlated strongly?

Where did material deviation occur?

Which Scenario Space lacked reality evidence?

How uncertain was the correlation claim?

Has the correlation changed since then?

Governance Boundary

SRCS begins when governed Simulation Outputs or Simulation Behaviors exist and corresponding Operational Reality evidence is sufficiently identifiable for comparison.

SRCS ends when: the comparison target is identified, the corresponding reality basis is established, relevant correlation criteria are applied, material deviations are identified, Reality Alignment is assessed, correlation uncertainty and limitations are preserved, and the Simulation-to-Reality relationship is expressed as a bounded, traceable correlation record.

SRCS does not determine: Operational Reality itself, universal simulation validity, general model integrity, final Validation Determination, deployment authorization, operational safety, or unrestricted reliance.

Relationship to ORA™

ORA™ governs Operational Reality.

It may establish: operational states, observed conditions, actual events, system configurations, real-world dependencies, or other reality references.

SRCS consumes the relevant governed reality basis and asks:

How does the simulation correspond to it?

Therefore:

ORA™ → Reality

SIMULOS® SRCS → Simulation ↔ Reality

Relationship Relationship to VALIDOS®

SRCS does not replace validation.

Its output may become important validation evidence.

VALIDOS® may subsequently assess: evidence fitness, source reliability, method, execution, determination, state, and reliance.

The architectural boundary is:

SRCS determines the governed correlation relationship.

VALIDOS® determines what validation conclusion that relationship can legitimately support.

Relationship to INTEGROS®

INTEGROS® may govern integrity of: Simulation Outputs, operational observations, comparison records, correlation artifacts, or preserved correlation history.

SRCS governs the meaning and boundaries of the correlation relationship itself.

Relationship to ADIT®

ADIT® may audit the evidence trail supporting Simulation-to-Reality Correlation.

SRCS remains the simulation-specific governance owner of the correlation relationship.

Relationship to SOGS

Simulation Output Governance Standard (SOGS) establishes:

what the simulation produced and what may legitimately be inferred directly from those outputs.

SRCS asks the next distinct question:

How does what the simulation produced correspond to Operational Reality?

Therefore:

Simulation Output ≠ Simulation-to-Reality Correlation.

Relationship to Reality Representation Standard

RRS governs how selected reality is represented inside the simulation.

SRCS provides an important downstream feedback mechanism:

Does that representation actually produce behavior corresponding sufficiently to Operational Reality?

Material deviation may indicate that Reality Representation requires reassessment.

Relationship to Simulation Assumptions

Simulation-to-Reality deviation may expose assumptions that: were inaccurate, became obsolete, apply only conditionally, or fail in particular Scenario Space.

SRCS identifies the correlation consequence.

SAS governs the assumption itself.

Relationship to Model & Environment Governance

Correlation deviation may arise from: model structure, environment configuration, parameters, versions, or boundary interactions.

SRCS identifies the mismatch.

SMES governs the relevant model and environment changes.

Relationship to Input & Data Governance

Input mismatch may affect correlation.

SRCS identifies the resulting Simulation-to-Reality difference.

SIDS governs the Simulation Input basis itself.

Relationship to Scenario Governance

Scenario Coverage limits Correlation Coverage.

A simulation cannot demonstrate Simulation-to-Reality Correlation in Scenario Space that has neither been adequately simulated nor observed in Operational Reality.

Relationship to Simulation Execution

Correlation relies upon traceable Simulation Execution.

If the execution that produced a Simulation Output cannot be reconstructed sufficiently, the correlation basis may itself become weakened or indeterminate.

Governance Outputs

SRCS may produce: Simulation-to-Reality Correlation Basis, Correlation Object Record, Reality Reference Record, Correlation Target, Comparison Context Record, Simulation–Reality Comparison Record, Correlation Criteria Record, Behavioral Correlation Record, State Correlation Record, Transition Correlation Record, Temporal Correlation Record, Spatial Correlation Record, Distributional Correlation Record, Sequence Correlation Record, Interaction Correlation Record, Boundary Correlation Record, Stress Correlation Record, Recovery Correlation Record, Local Correlation Record, Conditional Correlation Record, Partial Correlation Record, Correlation Profile, Correlation Strength Record, Simulation-to-Reality Deviation Record, Deviation Magnitude Record, Deviation Direction Record, Systematic Deviation Record, Localized Correlation Gap, Deviation Map, Reality Alignment Record, Reality Alignment Boundary, Correlation Coverage Record, Correlation Coverage Gap, Uncorrelatable Scenario Space Record, Correlation Uncertainty Record, Correlation Limitation Record, Correlation Change Record, Correlation Drift Record, Correlation Reassessment Trigger, Correlation Status, Simulation-to-Reality Traceability Map, and Governed Simulation-to-Reality Correlation Record.

Architectural Separation

SRCS must remain separated from four neighboring governance questions:

Reality Representation → How is reality represented inside the simulation?

Simulation Output Governance → What exactly did the simulation produce and what may be inferred from it?

Simulation-to-Reality Correlation → How does that simulated result correspond to actual Operational Reality?

Validation Governance → What formal validation conclusion may legitimately be established from the complete validation record?

This separation is fundamental to SIMULOS®.

Canonical Principle

Simulation similarity is not reality equivalence.

Supporting Principle

A simulation should be trusted as representative of reality only to the extent that the relevant relationship between simulated and observed reality has actually been examined, bounded, and kept traceable to the conditions under which that correspondence was established.

Foundational Principle

Simulation-to-Reality Correlation becomes methodologically legitimate only when similarity and deviation are evaluated against an identifiable Operational Reality basis, within explicit comparison conditions, using governed criteria, while preserving the uncertainty and boundaries of the resulting correlation claim.

Canonical Closing Statement

Simulation-to-Reality Correlation Standard establishes the governed bridge between Simulation and Operational Reality within SIMULOS®. By governing the identification of correlation targets, comparison conditions, correlation criteria, behavioral and state correspondence, transitions, distributions, timing, interactions, boundaries, deviations, Reality Alignment, Correlation Coverage, uncertainty, drift, reassessment, and complete Simulation-to-Reality traceability, SRCS prevents realistic appearance, numerical similarity, nominal agreement, or isolated successful comparison from being mistaken for Reality Equivalence. It enables simulations to state not merely that they resemble reality, but precisely where, under which conditions, to what extent, and with what uncertainty that correspondence has actually been demonstrated—while preserving the architectural boundary through which ORA™ governs Operational Reality and VALIDOS® governs the validation conclusions that may ultimately be drawn from the correlation record.

Module Architecture

→ [Simulation-Reality Comparison Module \(SRCM\)](#)

→ [Simulation Correlation Criteria Module \(SCCM\)](#)

→ [Simulation-Reality Deviation Module \(SDMM\)](#)

→ [Reality Alignment Module \(SRAM\)](#)

→ [Correlation Uncertainty Module \(SCUM\)](#)

Related Documents

→ [About Simulation-to-Reality Correlation Standard \(SRCS\)](#)

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